

CS161 Algorithms — Student Guide for This Semester

Based on the official course lecture schedule and resource page for this semester: <https://stanford-cs161.github.io/winter2026/lectures/>

How the Course Is Structured

The course runs 18 lectures, roughly two per week, moving from asymptotics and sorting, through trees, hashing, and graphs, into dynamic programming, greedy algorithms, MSTs, max-flow, and stable matching. Every lecture follows the same pattern of resources:

- **Pre-lecture exercise:** a short warm-up (sometimes with a notebook) to complete before class.
- **Lecture notes and slides:** the core content covered in that session.
- **Python notebook:** hands-on implementation of the algorithm just covered.
- **Concept check questions:** interactive and solved-PDF self-checks after lecture.
- **Additional readings:** assigned textbook sections (CLRS, “Algorithms Illuminated”) — required study, not optional extras.
- **Recording:** for review, or if a lecture is missed.

Grading

Component	Weight
Labs	45%
Midterm	15%
Final	40%
Attendance bonus	up to +10 points added to the Final

On attendance: those 10 points are real and can meaningfully change your final grade — don't treat them as trivial or skippable. But this bonus is earned by showing up across the semester, not requested afterward.

No extra-point requests will be considered at the end of the semester — if attendance wasn't tracked in your favor during the term, there is no way to make it up later. Show up consistently from week one.

Labs (45%): the single largest component. Rules and requirements for labs will be announced separately by our TA, **Ömer Kuru** — watch for that announcement and treat it as required reading, since it governs nearly half your grade.

A Warning Worth Taking Seriously

You cannot succeed in this course by studying $O(1)$ days before every exam. Exam questions will require real experience with the material and will not be straightforward — they will not reward last-minute cramming. You must keep up with the lectures and actively interact during class throughout the whole semester, so that those 3 hours each session are not wasted.

On AI usage: during exams, you will be on your own — no AI tools. For that reason, using AI to get answers to sample or study questions in advance is not recommended. Solve study questions as if you were already in the exam: on your own, under the same conditions, without looking up or generating the answer.

A Roadmap to Succeed

- **1. Never skip the pre-lecture exercise.** It's short by design and primes you for that day's material — lectures assume you've already wrestled with the setup.
- **2. Do the Python notebooks the same week they're posted.** Implementing an algorithm (e.g., QuickSort, Dijkstra, DP) cements it far better than reading slides alone.
- **3. Use the concept-check questions as a self-test.** Try them cold before checking the solved PDF, not just as a read-through.
- **4. Treat the additional readings as required, not optional.** CLRS and “Algorithms Illuminated” sections are assigned alongside each lecture for a reason — read them before or after class as directed, not just when something feels unclear.
- **5. Start labs early.** At 45% of your grade, falling behind here is the hardest thing to recover from — follow Ömer's lab guidelines from day one.
- **6. Attend every lecture you can.** Beyond the material, it directly banks toward your 10-point final bonus.
- **7. Review via recordings.** Use them to revisit tricky proofs (e.g., recurrences, SCC, max-flow correctness) before exams — not as a substitute for attending.
- **8. Build a weekly rhythm.** Pre-lecture exercise → assigned readings → attend lecture → notebook → concept check, all within the same week, so nothing piles up before the midterm or final.